REVIEW

HUMAN IMMUNODEFFICIENCY VIRUS: - THE ULTIMATE DETRIMENTAL TO HUMAN HEALTH

Abhipsa Chakraborty[1]

ABSTRACT

HIV belongs to lentivirus group of retrovirus family. The retrovirus are characterized by the possession of enzyme reverse transcriptase which allows the viral RNA to be transcripted into DNA and incorporated in host cell genome. In HIV structure there are two molecule of single standard DNA with protease, integrase, P24 core protein P17 matrix protein GP120 outer envelop glycoprotein CD4 receptor site and gp41 an transmembrane protein.

At the time of initial exposure the virus is transported by dendritic cell from mucosal surface to regional lymph nodes where the permanent infection is established. The HIV after entry into the body of human being enters into macrophages. Macrophages continued to produce HIV. HIV also destroys the T lymphocytes. The patient become immunodeficient. HIV transmits through sexual contact, contaminated blood and blood product, from infected mother to her child through placenta.HIV is now a manageable chronic condition but only for those who are aware of their diagnosis and who start antiretroviral therapy early enough.HIV infection is diagnosed either by the detection of virus specific antibodies or direct identification of viral material. There are some techniques like Detection of some IgG antibody to envelop components, IgG antibody to P24, Genome detection assays, Viral P24 antigen, Isolation of virus in culture. There are various effects of HIV infections like Neurological disease, Eye disorder, Gastrointestinal effect, Renal complications, Respiratory complications, Endocrine complications, Cardiac complications.Patients are monitored regularly. There are different types of monitoring. Immunological monitoring, Virological monitoring. The aim of the management of HIV infection to maintain physical and mental health, improve the quality of life,, increase survival rate. Restore and improve immune function, avoid onward transmission of virus. Different antiviral drugs. Reverse transcriptase inhibitors, protease inhibitors, integrase inhibitors are given to the patient. Patients should be monitored regularly. There may be drug resistance resulting from mutations in the protease or reverse transcriptase genes of the virus. Failure of antiretroviral treatment causes immunological deterioration. A changing therapy is needed. For different reasons the antiretroviral drugs may have to stopped. There are certain complication of antiretroviral therapy like allergic reactions, bone metabolism. There are some specific condition is associated with HiV infection. Thet are Pneumocystis, Toxoplasmosis, CMV retinities, lymphoma, cervical carcinoma, kaposis, sarcoma etc. High rate of HIV infection continue and prevention intervention remain fundamental to control the epidemic. Poverty, social unrest and war all contribute to the spread of HIV. Only clinical awareness of the people should prevent it to become a epidemic. Here, we review current methods of monitoring and control of HIV virus along with its effects . In addition we will discuss several inhibitors of HIV virus and detection of HIV virus along with investigation and tests of HIV virus.

KEYWORD:- Structure, pathogenesis, mode of infection, symptoms diagnosis, tests and monitoring therapy, effects of HIV, specific conditions associated with HIV infection, prevention and control.

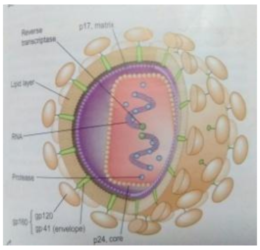
INTRODUCTION

HIV stands for human immunodeficiency virus. HIV belongs to the lentivirus group of the retrovirus family. There are two types HIV-1 and HIV-2. HIV-1 is the most frequently occurring strain globally. HIV-2 is almost entirely confined to West Africa, although there is some spread to Europe, particularly France and Portugal. HIV-2 has only 40% structural homology with HIV-1 and although it is associated with immunosuppression and AIDS, appears to take a more indolent course than HIV-1. Many of the drugs that are used in HIV-1 are ineffective in HIV-2. Retrovirus are characterized by the possession of the enzyme reverse transcriptase, which allows viral RNA to be transcribed into DNA and hence incorporated into the host cell genome. Reverse transcription is an error-prone process with a significant rate of mis-incorporation of bases. This, combined with a high rate of viral turnover, leads to considerable genetic variation and a diversity of viral subtypes or clades. On the basis of DNA sequencing, HIV-1 is divides into four district strains, which represent four independent cross species transfers, three(M,N and O) based on the chimpanzee related strains of SIV and 1(P) that may represent chimpanzee to gorilla to human transmission.

STRUCTURE OF HIV

HIV is a retrovirus that attacks helper T cells. The virus is a spherical enveloped virus of about 90-120nm in diameter. The envelope consists of a lipid bilayer derived from host cell membrane and projecting knob like glycoproteins spikes with pedicels formed of virus coded glycoproteins.

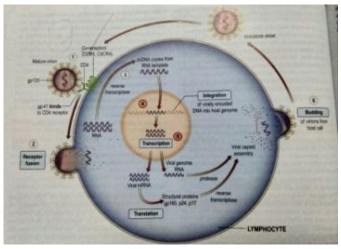
Two molecules of single-stranded RNA are shown within the nucleus. The reverse transcriptase polymerase converts viral RNA to DNA (a characteristic of retrovirus). The protease includes integrase (p32 and p10). The p24 (core protein) levels can be used to monitor HIV disease. P17 is the matrix protein; gp120 is the outer envelope glycoprotein which binds to cell surface CD4 molecules; gp41, a transmembrane protein, influences infectivity and cell fusion capacity.



The interrelationship between HIV and the host immune system is the basis of the parthogenesis of HIV disease. At the time of initial exposure virus is transported by dendritic cells from mucosal surfaces to regional lymph nodes where permanent infection is established. The host cellular receptor that is recognized by HIV surface glycoprotein gp120 is the CD4 molecule, which defines the cell populations that are susceptible to infection. The interaction between CD4 and HIV gp120 surface glycoprotein, together with host chemokine CCR5 co-receptor, is responsible for HIV entry into the cells. Although CCR5 CD4 memory T lymphocytes within all body systems are susceptible to infection and depletion those found in the gastrointestinal tract are heavily infected early in the process and become rapidly depleted leading to compromised mucosal immune function.

MODE OF ACTION OF HIV

On entering into the body of a person, HIV enters into macrophages, where RNA genome of the virus replicate to form viral DNA with the help of reverse transcriptase enzyme. This viral DNA gets incorporated into the host cell's DNA and directs the infected cells to produce more virus particles. The microphages continue to produce virus and in this way acts as a HIV factory. Simultaneously, HIV enters into helper T lymphocytes where it replicates and produces progeny viruses. The progeny viruses released in the blood attack other helper T lymphocytes. This is repeated leading to a progressive decrease in the number of helper T lymphocytes in the body of the infected person. During this period, the person suffers from bouts of fever, diarrhoea and weight loss. Due to decrease in the number of helper T lymphocytes, the person, starts suffering from infections of bacteria especially *Mycobacterium*, viruses, fungi and even parasites like *Toxoplasma*. The patient becomes so immunodeficient that he/she is unable to protect himself/herself against these infections. Besides helper T cells, HIV also infects macrophages and some B lymphocytes.



HIV is a virus that causes AIDS.

First description of AIDS in 1981 and the identification of the causative organism HIV in 1984, more than 20 million people have dies. At least 33 million people worldwide are living with HIV infection.

TRANSMISSION

AIDS is transmitted only when the blood of a patient comes in contact with the blood of a healthy person as in (i) Unprotected (without condom) sexual intercourse with an infected partner, which may cause tissue injury to permit blood contact. (ii) Transfusion of infected blood or blood product. (iii) Use of contaminated needles or syringes to inject drugs or vaccines. (iv) Use of contaminated razers for shaving. (v) Use of contaminated needle for boring pinnae. (vi) From organ transplant. (vii) From infected mother to child through placenta (viii) Artificial insemination.

INCUBATION PERIOD

The incubation period of AIDS ranges between 6 months to 10 years. Average period is about 28 months.

RISK GROUPS

These include homosexuals (infection through fragile rectal tissues during anal intercourse), intravenous drug abusers, recipients of transfused blood and blood products, haemophiliacs and heterosexual sex partners of AIDS patients.

SYMPTOMS

AIDS can manifest in two major ways: (i) malignant tumours in connective tissue, and (ii) viral, bacterial, protozoal and fungal infections of any system of the body. There is destruction of WBCs, damage to brain, fever, loss of appetite and weight over a short period, chronic diarrhoea, cough, night sweats, enlargement of lymph glands, lethargy, pharyngitis, nausea, headache, rashes etc.

REVIEW

DIAGNOSIS AND NATURAL HISTORY

HIV is now a manageable chronic condition, but only in those who are aware of their diagnosis and who start effective antiretroviral therapy early enough. Those newly diagnosed had a CD4 count below the threshold o start therapy (<350 cells/mm^3). The CD4 count becomes below 200 cells/mm^3 putting them at high risk of HIV associated pathology. Increasing the uptake of HIV testing is a major public health objective. Discussions about HIV testing and the consent required is straight forward and should be within the competencies of a wide range of healthcare professionals. Sensitive and specific point-of-care HIV antibody tests using either blood or oral fluids can give results within minutes and have extended the possibilities for diagnosis. All reactive point-of-care tests should be followed up with confirmatory serological assays, appropriate arrangements made to ensure patients receive test results and those who are found to be HIV positive have rapid routes into specialist care.

THE TEST

HIV infection is diagnosed either by the detection of virus specific antibodies (anti-HIV) or by direct identification of viral material. The recommended UK first lined assay is one which tests for HIV antibody AND p24 antigen simultaneously. These fourth generation assays have the advantage of reducing the time between infection and an HIV-positive test result to one month which is several weeks earlier than with sensitive third generation (antibody only detection) assays.

DETECTION OF IgG ANTIBODY TO ENVELOPE COMPONENTS

This is the most commonly used marker of infection. The routine tests used for screening are based on ELISA techniques, which may be confirmed with Western blot assays. Upto 3 months (mean 6 weeks) may elapse from initial infection to antibody detection (serological latency, or window period). These antibodies to HIV have no protective function and persist for life. As with all IgG antibodies, anti- HIV will cross the placenta. All the babies born to HIV-infected women will thus have the antibody at birth, In this situation, anti-HIV antibody is not a reliable marker of active infection and in uninfected babies will be gradually lost over the first 18 months of life. Simple and rapid HIV antibody assays are increasingly available, giving results within minutes. Assays that can utilize alternative body fluids to serum/plasma such as oral fluid, whole blood and urine are now available and home testing kits are being developed. These tests are extremely sensitive and may give false-positive results, making it necessary to perform a confirmatory test. A Serologic Testing Algorithm for recent HIV Seroconversions (STARHS) can be used to identify recently acquired infection. A highly sensitive ELISA that is able to detect HIV antibodies 6-8 weeks after infection is used on blood in patients with positive oral fluid test, in parallel with a less sensitive (detuned) test that identifies later HIV antibodies within 130 days.

IgG ANTIBODY TO p24 (ANTI-p24)

This can be detected from the earliest weeks of infection and through the asymptomatic phase. It is frequently lost as the disease progresses.

GENOME DETECTION ASSAYS

Nucleic acid-based assays that amplify and test for components of the HIV genome are available. These assays are used to aid diagnosis of HIV in babies of HIV-infected mothers or in situations where serological tests may be inadequate, such as in early infection when antibody may not be present, or in subtyping HIV variants for medicolegal reasons.

VIRAL p24 ANTIGEN (p24ag)

This is detectable shortly after infection but has usually disappeared by 8-10 weeks after exposure. It can be a useful marker in individuals who have been infected recently but have not had time to mount an antibody response.

ISOLATION OF VIRUS IN CULTURE

This is a specialized technique available in some laboratories to aid diagnosis and as a research tool.

EFFECTS OF HIV INFECTIONS

Eye disease

Eye pathology is usually seen in the later stages. The most serious is cytomegalovirus retinitis, which is sight-threatening. Retinal cotton wool spots due to HIV per se are rarely troublesome but they can be confused with CMV retinitis. Anterior uveitis can present as acute red eye associated with rifabutin therapy for mycobacterial infections in HIV. Steroids used topically are usually effective but modifications of the dose rifabutin is required to prevent relapse. Pneumocystis, toxoplasmosis, syphills and lymphoma can all affect the retina and the eye may be the site of first presentation.

Gastrointestinal effects

Weight loss and diarrhoea are common in untreated HIV-infected patients. Wasting is a common feature of advanced HIV infection, which although originally attributed to direct HIV effects on metabolism, is usually a consequence of anorexia. There is a small increase in resting energy expenditure in all stages of HIV, but weight and lean body mass usually remain normal during periods of clinical latency when the patient is eating normally. Gastrointestinal infections are common. An HIV enteropathy with varying degrees of villious atrophy has been described with chronic diarrhoea when no other pathogen has been found. Hypochlorhydria is reported in patients with advanced HIV disease and may have consequences for drug absorption and bacterial overgrowth in the gut. Rectal lymphoid tissue cells are the targets for HIV infection during penetrative anal sex and may be a reservoir for infection to spread through body.

RENAL COMPLICATIONS

Hiv-associated nephropathy (hivan)

It is although rare, can cause significant renal impairment, particularly in more advanced disease. It is most frequently seen in black male patients and can be exacerbated by heroine use.

Nephrotic syndrome

It is subsequent to focal glomerulosclerosis is the usual pathology, which may be consequence of HIV cytopathic effects on renal tubular epithelium. The course is usually relentlessly progressive and displays may be required.

Many nephrotoxic drugs

They are used in the management of HIV-associated pathology, particularly forcarnet, amphotericin B, pentamidine and sulfadiazine. Tenofovir is associated with Fanconi's syndrome.

Respiratory complications

The upper airway and lungs serve as a physical barrier to air-borne pathogens and any damage will decrease the efficiency of protection, leading to an increase in upper and lower respiratory tract infections. The sinus mucosa may also function abnormally in HIV infection and is frequently the site of chronic inflammation. Response to antibacterial therapy and tropical steroids is usual but some patients require surgical intervention. A similar process is seen in the middle ear, which can lead to chronic otitis media.

Lymphoid interstitial pneumonitis

It is well described in paediatric HIV infection but is uncommon in adults. There is an infiltration of lymphocytes, plasma cells and lymphoblasts in alveolar tissue. Epstein-Barr virus may be present. The patient presents with dyspnoea and a dry cough, which may be confused with pneumocystis infection. Reticular nodular shadowing is seen on chest X-ray. Therapy will steroids may produce clinical and histological benefit in some patients.

Endocrine complications

Various endocrine abnormalities have been reported, including reduced levels of testosterone and abnormal adrenal function. The latter assumes clinical significance in advanced disease when intercurrent infection superimposed upon borderline adrenal function precipitates clear adrenal insufficiency requiring replacement doses of glucocorticoids and mineralocorticoids. CMV is also implicated in adrenal-deficient states.

Cardiac complications

Cardiovascular pathology is increasingly recognized as a cause of morbidity in people with HIV. Although lipid dysregulation has been associated with antiretroviral medication (ARV), the observation has been made that HDL levels are lower in those with untreated HIV infection than in HIV-negative controls. In a large international study (SMART), ischaemic heart diseases was more common in those who took intermittent ARV therapy than in those who maintained viral suppression. Cardiomyopathy, although rare, is associated with HIV and may lead to congestive cardiac failure. Lymphocytic and necrotic myocarditis have been described. Ventricular biopsy should be performed to ensure other treatment causes of mycocarditis are excluded.

Neurological disease

Infection of the nervous tissue occurs at an early stage but clinical neurological involvement increases as HIV advances. This includes AIDS dermentia complex (ADC), sensory polyneuropathy and aseptic meningitis. These conditions are much less common since the introduction of HAART (highly active antiretroviral therapy). The pathogenesis is thought to be due both of the release of neurotoxic products by HIV itself and to cytokine abnormalities secondary to immune dysregulation. ADC has varying degrees of severity, ranging from mild memory impairement and poor concentration through to severe cognitive deficit, personality change and psychomotor slowing. Changes in affect are common and depressive or psychotic features may be present. The spinal cord may show vacuolar myelopathy histologically. In severe cases brain CT scan shows atrophic change of varying degrees. MRI changes consist of white matter lesions of increased density on T2-weighted sections. EEG shows non-specific changes consistent with encephalopathy. The CSF is usually normal, although the protein concentration may be raised. Patients with mild neurological dysfunction may be unduly sensitive to the effects of other insults such as fever, metabolic disturbance or psychotropic medication, any of which may lead to a marked deterioration in cognitive functioning.

Sensory polyneuropathy

It is seen in advanced HIV infection mainly in the legs and feet, although hands may be affected. Severe forms cause intense pain, usually in the feet, which disrupts sleep, impairs mobility and generally reduces the quality of life.

Autonomic neuropathy

It may also occur with postural hypotension and diarrhoea. Autonomic nerve damage is found in the small bowel. Didanosine and stavudine produce a similar neuropathy as a major toxic side-effect and are best avoided in patients with HIV neuropathy. HAART, using agents that penetrate the CNS can lead to significant improvements in cognitive function in many patients with ADC. It may also have a neuroprotective role.

INVESTIGATION AND MONITORING OF HIV-INFECTED PATIENTS

Initial assessment

Newly diagnosed patient should be reviewed by an HIV clinician within 2 weeks of diagnosis, or earlier if the patient is symptomatic or has other acute needs a full medical history, physical examination and laboratory evaluation should be undertaken in all newly diagnosed patients to determine the stage of infection, the presence of co-morbidities and co-infections and to assess overall physical, mental and sexual health. The initial assessment should also include details of socioeconomic situation, relationships, family and social support networks and substance misuse. Baseline investigations will depend on the clinical setting.

MONITORING

Immunological monitoring

Cd4 lymphocytes

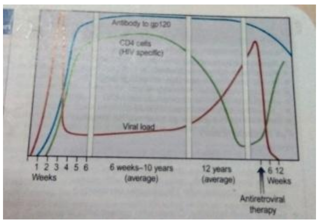
The absolute CD4 count and its percentage of total lymphocytes falls as HIV progresses. These figures bear a relationship to the risk of the occurrence of HIV-related pathology, with patients with counts below 200 cells at greatest risk. Rapidly falling CD4 counts and those below 350 are an indication for HAART. Factors other than HIV also affect CD4 number. CD4 counts are performed at approximately 3-monthly intervals unless values are approaching critical levels for intervention, in which case they are performed more frequently.

VIROLOGICAL MONITORING

Viral load (hiv rna)

HIV replicates at a high rate throughout the course of infection, with many billion new virus particles being produced daily. The rate of viral clearance is relatively constant in any individual and thus the level of viraemia is a reflection of the rate of virus replication. This has both prognostic and therapeutic value. The commonly used term 'viral load' encompasses viraemia and HIV RNA levels. Three HIV RNA assays for viral load are in current use:

- Branched-chain DNA (bDNA)
- Reverse transcription polymerase chain reaction (RT-PCR)
- Nucleic acid sequence-based amplification (NASBA)



Results are given in copies of viral RNA /ml of plasma, or converted to a logarithmic scale and there is good correlation between tests. The most sensitive test is able to detect as few as 20 copies of viral RNA/ml. Transient increases in viral load are seen following immunizations or during episodes of acute intercurrent infection and viral load measurements should not be carried out within a month of these events. By about 6 months after seroconversion to HIV, the viral set-point for an individual is established and there is a co-relation between HIV RNA levels and long-term prognosis, independent of the CD4 count. Those patients with a viral load consistently>100000 copies/ml have a 10 times higher risk for progression to AIDS over the ensuing 5 years than those consistently below 10000 copies/ml. HIV RNA is the standard marker of treatment efficacy. Both duration and magnitude of virus suppression are pointers to clinical outcome. None of the available therapies appears to be able to suppress viral replication indefinitely and a rising viral load, in a patient where compliance is assured, indicates drug failure. Various guidelines exist for viral load monitoring in clinical practice. Baseline measurement are followed by repeat estimations at intervals of 3-4 months, ideally in conjunction with CD4 counts to allow both pieces of evidence to be used together in decision-making. Following initiation of antiretroviral therapy or changes in therapy, effects on viral load should be seen by 4 weeks, reaching a maximum at 10-12 weeks, when repeat viral load should be carried out.

MANAGEMENT OF HIV-INFECTED PATIENT

Effective ART has transformed the management of HIV infection with a movement away from treating opportunistic conditions in immunosuppressed patients towards delivering long-term, effective suppressive therapy. With access and adherence to potent, tolerable antiretroviral drugs within a managed clinical setting, a 35-year-old initiating therapy in the UK in 2011 has a life expectancy of about 40 years. Nevertheless, there is still no cure for HIV and patients live with chronic, infectious and unpredictable condition. Limitations to efficacy include the inability of current drugs to clear HIV from certain intracellular pools, the occurrence of serious drug side-effects, adherence requirements, complex drug-drug interactions and the emergence of resistant viral strains. The aims of management in HIV infection are to maintain physical and mental health, improve the quality of life, increase survival rates, restore and improve immune function, avoid onward transmission of the virus and to provide appropriate palliative as needed. This requires long-term, maximal suppression of HIV activity using antiretroviral medication and management via a multidisciplinary team approach. Regular assessment is needed for details of intercurrent medical problems, medications, vaccinations, any recreational drug use, sexual history, reproductive decision making, cervical cytology, social situation to include support networks, employment, benefits and accommodation. Depression and anxiety are common among people living with HIV and mood and cognitive function should be routinely assessed. Psychological support may be needed not only for the patient but also for the family, friends and carers. Regular reviews of the sexual and reproductive health together with advice on reducing the risk of HIV transmission must be provided and future sexual practices discussed. Information to allow people to make informed choices about childbearing is required. The implications for existing family members should be considered and diagnostic testing offered as necessary. Regular monitoring of weight, body mass index, blood pressure and cardiovascular risk is required. Dietary assessment and advice should be freely accessible. General health promotion advice on smoking, alcohol, diet, drug misuse exercise should be given, particularly in the light of the cardiovascular, metabolic and hepatotoxicity risks associated with HIV and its treatment.

ANTIRETROVIRAL DRUGS (ART)

The treatment of HIV using antiretroviral therapy (HAART) continues to evolve and improve. Increased potency, reduced toxicity, greater convenience of formulation, compounds with different mechanisms of action coupled with improved understanding of drug resistance have combined to consistently improve HIV clinical outcome over time. An increase in the numbers of compounds, the array of drug-drug interactions, for example, combines to make HIV treatment complex and better clinical outcomes have been linked closely to physician expertise and the numbers of patients under direct care. Regularly updated treatment guidelines are produced in UK by British HIV Association.

REVERSE TRANSCRIPTASE INHIBITORS

Nucleoside/nucleotide analogye

Nucleoside reverse transcriptase inhibitors (NRTIs) inhibit the synthesis of DNA by reverse transcription and also act as DNA chain terminators. NRTIs need to be phosphorylated intracellularly for activity to occur. These were the first group of agents to be used against HIV, initially as monotherapy and later as dual drug combinations. Usually, two drugs of this class are combined to provide the backbone of a HAART regimen. Several fixed-closed NRTI combinations are available, which helps reduce the pill burden. NRTIs have been associated with mitochondrial toxicity, a consequence of their effect on the human mitochondrial DNA polymerase. Lactic acidosis is a recognized complication of this group of drugs. Nucleotide analogues have a similar mechanism of action but only require two intracellular phosphorylation steps for activity (as opposed to three steps for nucleoside analogues).

Non-nucleoside analogues

Non-nucleoside reverse transcriptase inhibitors (NNRTIs) interfere with reverse transcriptase by direct binding to enzyme. They are generally small molecules that are widely disseminated throughout the body and have a long half-life. NNRTIs affect cytochrome P450. They are ineffective against HIV-2. The level of cross-resistance across the class is very high. All have been associated with rashes and elevation of liver enzymes. Second-generation NNRTIs, such as etravirine and rilpivirine, with fewer adverse effects have some activity against virus resistant to other compounds of NNRTI class.

Protease inhibitors (pis)

These act competitively on the HIV aspartyl protease enzyme which is involved in the production of functional viral proteins and enzymes. As a consequence, viral maturation is impaired and immatured dysfunctional viral particles are produced. Most of the protease inhibitors are active at very low concentrations and *in vitro* are found to have synergy with reverse transcriptase inhibitors. However there are difference in toxicity, pharmacokinetics, resistance patterns and also cost, which influence prescribing. Cross-resistance can occur across the PI group. There appears to be no activity against human aspartyl proteases, although there are clinically significant interactions with the cytochrome P450 system. This is used to therapeutic advantage, boosting blood levels of PI by blocking drug breakdown with small doses of ritonavir. PIs have been linked with abnormalities of fat metabolism and control of blood sugar and some have been associated with deterioration in clotting function in people with haemophilia. Second generation PIs (darunavir, tipranavir) with activity against viruses resistant to the first generation drugs are available.

Integrase inhibitors

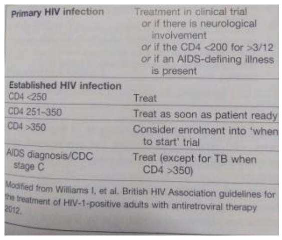
These drugs act as a selective inhibitor of HIV integrase, which blocks viral replications by preventing insertion of HIV DNA into the human DNA genome. Three compounds have so far been developed, although only one, raltegravir is used. Raltegravir is metabolised by glucuronidation and does not require retroviral drug boosting. It is effective in treatment of both experienced and naive patients.

Co-receptor blockers

Maraviroc is a chemokine receptor antagonist which blocks the cellular CCR5 receptor entry by CCR5 tropic strains of HIV. These strains are found in earlier HIV by infection and, with time adaptations allow the CXCR4 receptor to become the more dominant form. The drug is metabolized by CYP p450 (3A), giving the potential for drug-drug interactions. Tropism assays to establish that the patient is carrying a CCR5 tropic virus are required before treatment is used.

Fusion inhibitors

Enfurvitide is the only licensed compound in this class of agents. It is an injectible peptide derived from HIV gp41 that inhibits gp41-mediated fusion of HIV with the target cell. It is synergistic with NRTIs and PIs. Although resistance to enfurvitide has been described, there is no evidence of cross-resistance with other drug classes. As it has an extracellular mode of action there are few drug-drug interactions. Side-effects relate to the subcutaneous route of administration in the form of injection site reactions.



Primary HIV infection	Treatment in clinical trial *or* if there is neurological involvement *or* if the CD4 <200 for >3/12 *or* if an AIDS-defining illness is present
Established HIV infection	
CD4 <250	Treat
CD4 251–350	Treat as soon as patient ready
CD4 >350	Consider enrolment into 'when to start' trial
AIDS diagnosis/CDC stage C	Treat (except for TB when CD4 >350)

Modified from Williams I, et al. British HIV Association guidelines for the treatment of HIV-1-positive adults with antiretroviral therapy 2012.

STARTING THERAPY

Although the benefits of HAART in HIV infection are now indisputable, treatment regimens require a long-term commitment to high levels of adherence. Risks of therapy include short- and longer-term side-effects, drug-drug interactions and the potential for development of resistant viral strains. The full involvement of patients in therapeutic decision making is essential for success. Various national guidelines and treatment frameworks exist. Laboratory marker data, including viral load and CD4 counts, together with individual circumstances should guide therapeutic decision making. Questions still remain about the best time to start therapy. Clear clinical benefit has been demonstrated with the use of antiretroviral drugs in advanced HIV disease. All patients will symptomatic HIV disease; AIDS or CD4 count that is consistently below 200 cells/mm^3 should initiate treatment as soon as possible. In such situations, there is a significant risk of serious HIV-associated morbidity and mortality and the longer-term prognosis for patient initiating therapy below 200 CD4 cells/mm^3 is not as good as for those who start at higher counts.

CHOICE OF DRUG

The drug regimen used for starting therapy must be individualized to suit each patient's needs. As differences in drug efficacy become less marked, fitting the drugs to the patient's need and lifestyle are key to success. Treatment is initiated with three drugs, two NRTIs in combination, with either an NNRTI or a boosted protease inhibitor. The development of fixed-dose coformulations reduces pill burden, increases convenience and facilitates adherence. New drug classes such as integrase inhibitors and regimens without an NRTI backbone are still in the trial process and are likely to be the next development in first-line therapy.

Nucleoside reverse transcriptase inhibitor (nrti)

Two NRTIs form the backbone is influenced by efficacy, toxicity and ease of administration. The availability of once daily used one-tablet fixed-dose combinations, Truvada and Kivexa has led the majority of patients who are naive to medication being prescribed one of these as their 2NRTI backbone. Kivexa should only be used in those who are HLAB*5701 negative. Combivir has lower efficacy than Truveda, a twice-daily dosing schedule and poorer toxicity profile. Data comparing Truveda and Kivexa in naive patients has demonstrated noninferiority of Kivexa at viral levels below 100000 copies/ml and there is concern over an association of abacavir with cardiovascular events.

Non-nucleoside reverse transcriptase inhibitor (nnrti)

The decision about use of NNRTI or boosted PI will depend on the particular circumstances of each patient but in the UK, an NNRTI-based regimen is most commonly prescribed to naive patients. Efavirenz is the recommended option in the UK, having demonstrated good durability over time, potency at low CD4 counts and in high viral loads. Efavirenz has the advantage of once-daily dosing but is associated with CNS side-effects such as dysphoria and insomnia and is contraindicated in pregnancy. The fixed-dose preparation of efavirenz co-formulated with Truvada allows for a 'one pill once a day' regimen.

Protease inhibitors

This class of drugs has demonstrated excellent efficacy in clinical practice. PLs are usually combined with a low dose of ritonavir which provides a pharmacokinetic advantage by blocking cytochrome P450 metabolism. Using this approach, the half life of active drug is increased, allowing greater drug exposure, fewer pills, enhanced potency and the risk of resistance minimized. Atazanavir, darunavir or lopinavir, boosted with ritonavir, are most commonly used as first line therapy.

MONITORING THERAPY

Success rates for initial therapy using modern ARVs judged by virological response are very high. By 4 weeks of therapy, the viral load should have dropped by atleast 1 log10 copies/ml and by 12-24 weeks should be below 50 copies/ml. A suboptimal response at either time point demands a full assessment and possible change in therapy. Once stable on therapy the viral load should be routinely measured every 3-4 monthsCD4count should be repeated at one and three months . After starting HAART and then every 3-4 months. Once both the viral load is below 50 copies/ml and the CD4 count has been above 350 cells for at least 12 months frequency may fall to 6 monthly. Impaired immunological recovery is associated with treatment initiation in advanced infection (a low CD4 count and late presentation) and with older age. Regular clinical assessment should include review of adherence to and tolerability of the regimen, weight, blood pressure and urinalysis. Patients should be monitored for drug toxicity, including full blood count, liver and renal function and fasting lipids and glucose level.

Drug resistance

Resistance to ARVs results from mutations in the protease reverse transcriptase and intergrase genes of the virus. HIV has a rapid turnover with 10^8 replications occurring per day. The error rate is high, resulting in genetic diversity within the population of virus in an individual, which will include drug- resistant mutants. When drugs only partially inhibit virus replication there will be selection pressure for the emergence of drug resistant strains. The rate at which the resistance develops depends on the frequency of pre-existing variants and the number of mutations required. Resistance to most NRTIs and PIs occur with an accumulation of mutations whilst a single point mutation will confer high-level resistance to NNRTIs. There is evidence for the transmission of HIV strains that are resistant to all or some classes of drugs. Studies of primary HIV infection have shown prevalence rates between 2% and 20%. Prevalence of primary mutations associated with drug resistance in chronically infected patients not on treatment ranges from 3% to 10% in various studies.

Drug interaction

Drug therapy in HIV is complex and the potential for clinically relevant drug interactions is substantial. Both PI and NRTIs are able to variably inhibit and include cytochrome P450, influencing both their own and other metabolic rates. Both inducers and inhibitors of cytochrome P450 are sometimes prescribed simultaneously. Induction of metabolism may result in sub therapeutic antiretroviral drug level with the risks of treatment failure and development of viral resistance, whilst inhibition can raise drug levels to toxic values and precipitate adverse information. Conventional and complementary therapies affect cytochrome P450 activity and may precipitate substantial drug interaction. Therapeutic drug monitoring indicating peak and trough plasma levels may be useful in certain settings.

Adherence

Patient's beliefs about their personal need for medicines and their concerns about treatment affect how and whether they take them. Adherence to treatment is pivotal to success.. Levels of adherence below 95% have been associated with poor virological and immunological responses although some of the newer ARVs are more forgiving. Poor absorption and low bioavailability mean that for some compounds trough levels are barely adequate to suppress viral replication and missing even a single dose will result in plasma drug levels falling dangerously low. Factors implicated in poor adherence may be associated with the medication, with the patient or with the provider. Patient factors include the level of motivation and commitment to the therapy, psychological wellbeing, the level of available family and social support and health beliefs. Supporting adherence is a key part of clinical care and specific guidelines are available. Education of patients about their condition and treatment is a fundamental requirement for good adherence, as is education of clinicians in adherence support techniques.

Treatment failure

Failure of antiretroviral treatment, i.e. persistent viral replication causing immunological deterioration and eventual clinical evidence of disease progression, is caused by a variety of factors, e.g. poor adherence, limited drug potency and food or other medication may compromise drug absorption. There may be drug interaction or limited penetration of drug into sanctuary sites such as the CNS, permitting viral replications. Side-effects and other patient-related elements contribute to poor adherence.

CHANGING THERAPY

A rise in viral load, falling CD4 count or new clinical events that imply progression of HIV disease are all reasons to review therapy. Reasons for treatment failure include the emergence of resistant viral strains, poor patient adherence or intolerance/adverse drug reactions. Virological failure, two consecutive viral loads of >400 copies/ml in previously fully suppressed patient, requires investigations. Viral genotyping should be used to help select future therapy, choosing at least two new agents to which the virus is fully sensitive. If a new suitable treatment option is available it should be started as soon as possible.

STOPPING THERAPY

Antiretroviral drugs may have to be stopped in, for example, cumulative toxicity, or potential drug interactions with medications needed to deal with another more pressing problem. If adherence is poor, stopping completely may be preferable to continuing with inadequate dosing, in order to reduce the development of viral resistance. Poor quality of life and the view of the patient should be discussed. NNRTIs efavirenz and nevirapine have no long half-lives and therefore should be stopped before the other drugs in the mixture to reduce the risk of drug resistance. If this is not possible lopinavir/ritonavir may be used either in substitution of the NNRTIs or as monotherapy or several weeks to cover the period of sub therapeutic levels.

COMPLICATIONS OF ANTIRETROVIRAL THERAPY

Allergic reactions

Allergic reactions occur with greater frequency in HIV infection and have been documented with all the antiretroviral drugs. Abacavir is associated with potentially fatal hypersensitivity reactions. There may be a discrete rash and often a fever coupled with general malaise and gastrointestinal and respiratory symptoms resolve when abacavir is withdrawn. Re-challenge with abacavir can be fatal and is contraindicated.

Lipodystrophy and metabolic syndrome

A syndrome of lipodystrophy occurs in patients with HIV on HAART comprising characteristic morphological changes and metabolic abnormalities.

Mitochondrial toxicity and lactic acidosis

Mitochondrial toxicity, mostly involving the older drugs stavudine and didanosine in nucleoside analogue class, leads to raised lactate and lactic acidosis, which has in some cases been fatal.

Bone metabolism

A variety of bone disorders have been reported in HIV, in particular osteopenia, osteoporosis and avascular necrosis.

Pregnancy

Management of HIV positive pregnant women requires close collaboration between obstetric, medical and paediatric teams. The management aim is to deliver a healthy, uninfected baby to a healthy mother without prejudicing the future treatment options of the mother. HIV-positive women are advised against breast-feeding, which doubles the risk of vertical transmission. Delivery by caesarean section reduced the risk of vertical transmission in the pre HAART era but if the women is on effective HAART and the labour is uncomplicated vaginal delivery carries no additional risk. For women eligible for treatment of their own HIV disease, whether pregnant or not, triple therapy is the regimen of choice. Risk of vertical transmission increases with viral load. Although the fetus will expose to more drugs, the chances of reducing the viral load and hence preventing infection are greatest with a potent triple therapy regimen in the mother.

Treatment

Treatment should start as soon as possible and continue during delivery. The baby should receive zidovudine for 4 weeks postpartum and the mother remain on ARVs with appropriate monitoring and support. Women who do not need treatment for themselves should be prescribed a short course of antiretroviral therapy initiated at approximately 20 weeks of pregnancy to reduce vertical transmission. Efavirenz has been associated with developmental abnormalities in primate models and several retrospective cases of neural tube defects in babies born to women taking efavirenz have been reported. Women who have conceived on the drug should discuss the risks and benefits of continuing or switching with an expert clinician.

Post-exposure prophylaxis

Healthcare workers may be treated following occupational exposure to HIV as may those exposed sexually. The risk of acquisition of HIV following exposure is dependent upon the risk that the source is HIV positive (if this is unknown in a sexual exposure. PEP may be useful upto 72 hours after possible exposure. UK, the standard regimen is Truvada plus Kaletra although this may be varied depending on what is known about the source. Treatment is given for 4 weeks and the recipient should be monitored for toxicity.

Opportunistic infections in the HAART era

Although effective antiretroviral therapy has resulted in a remarkable decline in opportunistic infections in patients with HIV, not all those at risk may be either on or adhering to effective treatment. The types of infections seen in the context of HIV have altered with fewer episodes of the ‘classic’ Ols, such as pneumocystis pneumonia and cytomegalovirus, but an increase in community-acquired infections such as *Strep.pneumoniae* and *Haemophilus influenza*. Immune reconstitution with HAART may produce unusual responses to opportunistic pathogens and confuse the clinical picture.

PREVENTION OF OPPORTUNISTIC INFECTION IN HIV-INFECTED PATIENTS

Avoid infection

Exposure to certain organisms can be avoided in those known to be HIV-infected. Attention to food hygiene will reduce exposure to Salmonella, toxoplasmosis and Cryptosporidium and protected sexual intercourse will reduce exposure to herpes simplex virus, hepatitis B and C and papillomaviruses. Cytomegalovirus negative patients should be given CMV-negative blood products. Travel-related infection can be minimized with appropriate advice.

Immunization strategies

Immunization may not be as effective in HIV-infected individuals. Hepatitis A and B vaccines should be given for those without natural immunity who are at risk, particularly if there is co-existing liver pathology.

FUNGAL INFECTIONS

Pneumocystis jiroveci

This organism most commonly causes pneumonia but can cause disseminated infection. It is not usually seen until patients are severely immunocompromise with a CD4 count below 200. This introduction of effective antiretroviral therapy and primary pophylaxis in patients with CD4 <200 has significantly reduced the incidence in the UK. The organism damages alveolar epithelium, which impedes gas exchange and reduces lung compliance.

PROTOZOAN INFECTION

Toxoplasmosis

Toxoplasmosis gondii most commonly causes encephalitis and cerebral abscess in the contest of HIV, usually as a result of reactivation of previously acquired infection.

MICROPORIDIOSIS

Enterocytozoon and Septata are a cause of diarrhoea. Spores can be detected in stools using a trichrome or fluorescent stain that attaches to the chitin of spore surface. HAART or immune restoration is the treatment of choice and can have a dramatic effect.

VIRAL INFECTIONS

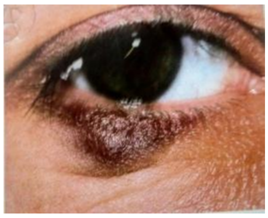
CMV retinitis

This occurs once the CD4 count is below 50. Although usually unilateral to begin with, the infection may progress to involve both eyes. Presenting features depend on the areas of retina involved and include floaters, loss of visual acuity, field loss and scotomata, orbital pain and headache.

INFECTIONS DUE TO OTHER ORGANISMS

Kaposi's sarcoma

Kaeposi's sarcoma in association with HIV behaves more aggressively than that associated with HIV-negative populations.



Lymphoma

A significant proportion of patients with HIV develop lymphoma, mostly of the non-Hodgkin's, large B-cell type.

Cervical carcinoma

Women with HIV are at increased risk of cervical cancer caused by oncogenic subtypes of human papillomarvirus. Annual cervical cytology is indicated to monitor for pre-malignant changes.

CONCLUSION

Prevention and control

High rates of new HIV infection continue and prevention interventions remain fundamental to the control of epidemic. Combination of behavioural, biomedical and structural approach with interventions appropriate for the particular population at risk is required. Vaccine development has been hampered for genetic variability of the virus and the complex immune response that is required from the host with disappointing results from trials of candidate agents. Consistent condom use and education for behaviour change have been key strategies. Provision of clean injecting equipment has been successful in those countries where it has been implemented.

ACKNOWLEDGEMENT

The author would like to thank Dr Amar Nath Chakraborty.

BIBLIOGRAPHY AND REFERENCE

John TJ, Dandona L, Sharma VP, Kakkar M. Continuing challenge of infectious disease in India. Lancet 2011: 377: 252-269.

Barnett T, Whiteside A, AIDS in the Twenty-First Century, Disease and Globalisation, 2nd edition. London: Palgrave; 2006.

Oxford Handbook of Clinical Medicine, South Asia edition, Murray Longman and Andrew Baldwin.

CDC. Revised recommendation for HIV testing of adults, adolescents and pregnant women in healthcare settings, MMWR Morb Mortal Wkly Rep 2006; 55:1-17.

Comprehensive Biology, Laxmi publication private limited, by Dr J.P Sharma.

Chadwick DR, Geretti AM. Immunization of the HIV infected traveller. AIDS 2007; 21:787-794.

REVIEW ARTICLE

Name:- Abhipsa Chakraborty

Enrolment number:- A91704118039

Course:- Term Paper

Program:- BTech Biotechnology

Course code:- BTTP

Year:- 2018-2022

Section:- A

Semester:- 3

Topic:- HIV and AIDS